

GIACOMO GUARESI

SOFTWARE · HARDWARE · INDUSTRY

MILAN AREA, ITALY

I build the software that runs machines, and the electronics it runs on: firmware, boards, control stacks and the backends around them – from schematic to production. Eight years in industry, from regulated pharmaceutical plants to additive manufacturing. Looking for development and R&D roles, remote or hybrid.

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.01 STACK

ELECTRONICS	Schematic design · PCB routing · Design for production · EasyEDA · <i>boards designed and shipped inside a commercial machine</i>
EMBEDDED	C · C++ (C++11/17/20) · STM32 · ESP32 · Arduino · CAN bus · Sensor and actuator control · Embedded Linux (ARM) · Custom kernel builds
INDUSTRIAL	SCADA (iFIX, Movicon) · Modbus · OPC-UA · Profinet · EtherCAT · MQTT · Environmental monitoring · GMP and ISO 9001 regulated environments
ADDITIVE	Klipper · Mainsail · OrcaSlicer · Slicing and G-code · Machine control stacks · Upstream open-source contribution
BACKEND	C# / .NET (Framework & Core) · Python · Node.js · Spring · REST API design · OAuth · Kafka · Redis pub/sub · Monolith, microservices, event-driven
DATA	PostgreSQL · MySQL · Oracle · MongoDB · Redis · SQLite · Process historians · Datasets up to ~1 TB · Query tuning and indexing
SYSTEMS	Linux end to end (server, systemd, networking, scripting, custom kernels) · Docker · Docker Compose · GitHub Actions · GitLab CI · Advanced Git · Production deploys, domains, TLS
FRONTEND	React · Angular · TypeScript · HTML · CSS – <i>a supporting skill, roughly a fifth of my work</i>
AI IN PRODUCTION	LLM integration with function calling (Gemini) · OpenCV · TensorFlow · Defect detection · Predictive maintenance · Adaptive thresholds, regression and clustering on process data
LEARNING NOW	KiCad · Kubernetes · Rust · Agentic AI systems

.02 WORK

2024 – NOW	GINGER ADDITIVE
Additive	<i>Software Engineer – software, electronics and IT across the company</i> → A startup building industrial pellet-fed 3D printers: I have followed the G1 from the earliest prototypes to units sold on the market. → Machine firmware and control stack on a Klipper base, over custom Linux images with a purpose-built kernel. → Several of the printer's PCBs designed from schematic to routing to production, along with their firmware. → A dedicated slicer built on OrcaSlicer around pellet extrusion; changes pushed upstream to OrcaSlicer, Klipper and Mainsail rather than into a private fork. → Around ten internal applications for running the company, plus the servers, deployments, CI pipelines and infrastructure underneath. → An internal AI assistant wired into the project management system (Gemini with function calling), in daily use.
2023 – 2024	INTRE
Consulting	<i>Full Stack Developer – enterprise consulting</i> → Projects for large industrial and professional-services organisations; details under NDA. → React, Angular and TypeScript front ends over containerised, cloud-native back ends running on managed Kubernetes.

2019 – 2023

Pharmaceutical

AM INSTRUMENTS

SCADA Programmer & R&D

- Four years on the plant floor of major Italian pharmaceutical manufacturers, building and integrating environmental monitoring systems in iFIX and Movicon under GMP and ISO 9001.
- Hundreds of sensors and legacy machines integrated over Modbus, OPC-UA, Profinet and EtherCAT; process historians approaching a terabyte, with query tuning and indexing.
- R&D on a hydrogen-peroxide room sterilisation system, built across its whole stack: embedded C++ on an ARM controller running embedded Linux, CAN bus control of the sensors and the dosing, a .NET web portal, iOS and Android apps, and Python tooling.
- Defect detection, predictive maintenance and adaptive thresholds applied to the process data (OpenCV, TensorFlow).

2018 – 2019

On client sites

INTERLEM

Full Stack Developer – on site at client companies

- First job, hired straight after my diploma by the company where I had done my school placement.
- Industrial IoT product: smart scales measuring how much gas is left in welding cylinders, Arduino and LoRa hardware built in a team, with a Spring and PostgreSQL management portal.
- Full refactor of a legacy Delphi business portal, maintained until its .NET replacement shipped.

.03 EDUCATION & LANGUAGES

DIPLOMA

Perito Informatico (technical diploma in computer science) – ITIS Henseberger, Monza, 2018. *No degree and no certificates: everything since has been self-taught and learned on the job.*

LANGUAGES

Italian native · English solid

OWN PROJECTS

FAInder, an app and scraper for the FAI heritage sites across Italy · A hiking and alpinism notes site with a system for keeping the printed pages up to date

HOW I WORK

I use AI every day as a tool: most of the code I ship passes through it, and I read every line before it goes out.

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